Academic Paper Reviewer: A Multi-Perspective Review Skill for Computer Vision and Machine Learning Papers

Technical Report, version 2.10.1

Academic Paper Reviewer project
https://github.com/Shyandram/academic-paper-reviewer.skill

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Abstract

We describe *Academic Paper Reviewer*, a Markdown skill for Claude and Codex agents that simulates an editorial process for a manuscript. The process has a field-aware panel of five independent reviewers, a literature check that cannot find the paper under review, a report check, and an Editor-in-Chief (EIC) who weights each review by its quality and makes the decision. Scores are aggregated by a deterministic, standard-library Python script and translated into the forms of 11 venues.

We evaluate the skill in two ways.

- **Accepted papers with planted flaws.** On 30 recently accepted CVPR/ECCV/ICLR papers, 96% of cited locations, 99% of quoted phrases and all but two of 842 quoted numbers are found in the paper, and the reviewers mention all six planted fatal flaws.
- **ICLR 2026 calibration.** This follows the paperreview.ai protocol. On the balanced 86-paper set with Sonnet at high effort, the reviewers' mean rating reaches Spearman 0.47 [0.27, 0.64] with the mean human rating (two human reviewers of the same papers: 0.14) and an acceptance AUC of 0.64. On the ICLR scale the normalized mean absolute error is 0.084, with essentially no bias (NME −0.003). The 20-paper pilot gave the same ranking (0.48).
- **Issue coverage.** On the 86 papers the panel raises 46% of the human reviewers' main issues and 74% of those two or more reviewers share; the more important the issue (consensus, a reject vote, a reviewer's lead point), the more often it is raised. On a 20-paper pilot, an optional Gap Scout that names concrete missing items raises 65% alone and 73% with the panel. A redesign that moved these ideas into the panel (version 2.11) raised no more issues, ranked papers worse (Spearman 0.32) and cost 3.5× as much, so it was withdrawn.

A smaller model (Haiku) shows no reliable signal. We also report a negative result: NMAE alone does not measure skill on this pool, because a constant predictor scores 0.089. The main remaining weakness is compression: 78% of papers receive the same final rating.

1 Introduction

Large language models can now write fluent paper reviews. Fluency, however, is not the goal of a review. Authors need the decisive weaknesses found, grounded in the manuscript, and ranked by how much they matter at a target venue. Three failure modes are common:

- **Ungrounded criticism:** weaknesses that the paper already answers, or locations and numbers that do not exist.
- **Compressed scores:** every paper rated "borderline".
- **Leakage:** the reviewer finds the paper's own camera-ready version, reviews or acceptance status, through retrieval or memory.

Prior AI reviewers either prompt a general model once per paper, as in the AI Scientist [4], or fine-tune an open model on public reviews: OpenReviewer [2], SEA-E [8], CycleReviewer [7] and DeepReviewer [12]. Academic Paper Reviewer instead addresses these failure modes with structure rather than with a single prompt or a single trained model.

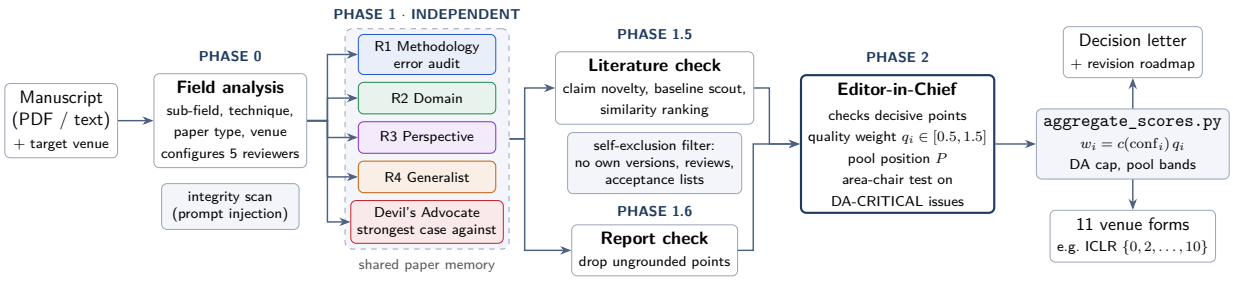


Figure 1: Full-mode workflow. Five reviewers write independently from a shared factual paper memory. A literature check, which cannot retrieve the paper under review, and a report check run before the Editor-in-Chief. The EIC weights each review by quality and checks decisive points against the manuscript. A deterministic script aggregates scores and translates them into each venue's form.

- Several reviewers with distinct roles write independently.
- A Devil's Advocate attacks the paper.
- A report check removes ungrounded points.
- An EIC acts as area chair.

The skill is tuned for computer vision, deep learning and ML papers, with task-specific checklists for detection, segmentation, 3D/NeRF/3DGS, restoration, generation, VLMs, video, medical imaging and efficient models, but it can be used in any field. Its output is a *simulation* to help authors improve a paper, not a prediction of a venue's decision.

Contributions.

1. A review workflow with independent role-based reviewers, a quality-weighted EIC, and deterministic score aggregation into venue-specific forms (Section 2).
2. Safeguards against prompt injection and self-leakage in literature search (Section 3).
3. A reproducible calibration harness following the paperreview.ai protocol [5], extended with CSPaper-style error metrics [6] and a no-split pilot mode (Section 5).
4. Pilot and full-evaluation results, including what the skill did badly and the fixes that followed.

2 System design

2.1 Workflow

A full review runs the phases in Figure 1. The phases are Markdown instructions in `SKILL.md` and `agents/*.md`; scripts are optional helpers.

Phase 0: field analysis. A field analyst identifies the paper's sub-field, technique, paper type (empirical, theoretical, dataset, system, benchmark, analysis, position) and target venue, and configures five reviewers with specific expertise. An example is "token pruning + latency benchmarking" rather than "deep learning". `integrity_scan.py` flags text addressed to reviewers or AI systems, as well as hidden characters.

Phase 1: independent reviews. Five reports are written independently: Methodology (R1), Domain (R2), Perspective (R3), Generalist (R4) and a Devil's Advocate (DA). In Claude Code they can run as parallel subagents. The Methodology reviewer runs an error audit in nine categories (result inconsistencies, equation and proof errors, assumptions, protocol flaws, claim distortion, inconsistent definitions, citation errors, omissions, figure/table integrity). It works from a shared factual *paper memory*, and every error needs a location, a verbatim quote and a concrete check.

Phase 1.5: literature check. This runs only with network access and after the reports are drafted. Searches are driven by contribution claims and benchmarks: a novelty verdict per

Table 1: Paper-type weights of the seven 0–100 dimensions (each row sums to 1).

Type	Rigor	Clarity	Novelty	Signif.	Exp. val.	Repro.	Ethics
empirical	.20	.10	.15	.15	.20	.10	.10
theoretical	.30	.10	.20	.15	.10	.05	.10
dataset	.15	.10	.10	.20	.15	.15	.15
system	.20	.10	.10	.20	.20	.10	.10
benchmark	.15	.15	.10	.20	.20	.15	.05
analysis	.25	.10	.20	.20	.15	.05	.05
position	.15	.20	.25	.25	.05	.00	.10

claim, a baseline scout, a coverage pass, a field snapshot and a similarity ranking. Section 3 describes how the paper itself is excluded.

Phase 1.6: report check. Before the EIC reads anything, vague weaknesses, points the paper already answers, and claims the paper contradicts are removed. Cosmetic issues are grouped as Minor.

Phase 2: EIC meta-review. The EIC reads every report and checks the decisive points against the paper. It assigns each reviewer a quality weight in $[0.5, 1.5]$, states its own pool position, and decides. Each DA-CRITICAL issue gets an area-chair test ("would I reject on this alone?").

Other modes reuse the same components: quick assessment, methodology focus, guided (Socratic) review, rebuttal drafting and re-review of a revision.

Post-review analysis (optional). After the review, or on its own, the user can ask for a *Gap Scout*: a separate agent that reads only the redacted paper and lists 20–30 concrete concerns reviewers are likely to raise (the specific missing experiment, baseline, definition or setting, with a location). It gives no scores, is never called by default and never changes the review; its list is a checklist for the authors (Section 5.6).

2.2 Scores

Every non-DA reviewer gives NeurIPS-style 1–4 ratings, an Overall 1–6 recommendation (6 Strong Accept … 1 Strong Reject), a confidence 1–5, and seven 0–100 dimension scores: technical rigor, clarity, novelty, significance, experimental validation, reproducibility and ethics. The DA gives the 0–100 set only. Dimensions are combined with paper-type weights (Table 1).

Two score lists are kept. The **official** list holds the reviewers' raw ratings and their plain means, which is what authors see; it is never modified. The **EIC assessment** weights reviewer i by

$$w_i = c(\text{conf}_i) \cdot q_i, \qquad c(5..1) = (1.00, 0.85, 0.70, 0.55, 0.40), \quad q_i \in [0.5, 1.5], \tag{1}$$

where q_i is the EIC's quality weight. The decision starts from the EIC-weighted mean of the Overall ratings, rounded half up. The weighted 0–100 composite is a cross-check: it maps to Overall at thresholds 85/70/60/50/35. Two further rules apply:

- **DA cap.** Upheld DA-CRITICAL issues cap the decision at 4 if they can be resolved by clarification and at 3 otherwise.
- **Consistency warnings** flag disagreements that the EIC must explain.

Calibration to the submission pool (2.10). Each reviewer and the EIC first state a *pool position* P: the share of the venue's submissions the paper is better than, using real base rates (about 25% acceptance at major ML/CV conferences; ICLR 2026 about 27% accepted, 45% rejected, 24% withdrawn, 5% desk rejected). They then score within the matching band:

Table 2: Grounding on 30 accepted papers: share of cited items not found in the paper. Numbers the reviewers recomputed themselves (525) are excluded and were spot-checked by hand.

Check	Checked	Not found	Rate
Cited locations (tables, figures, sections, equations)	1101	48	4%
Quoted phrases	778	9	1%
Quoted numbers	842	2	0%

P	≥ 95	85–94	70–84	55–69	30–54	< 30
composite	85+	73–84	62–72	55–61	42–54	< 42

A composite of 62 is the accept-side bar. Uncertainty is never a reason to score a paper as borderline. The aggregation script warns when a composite lies more than 5 points outside its pool band, and when all reviewers sit in the borderline band.

2.3 Venue forms

Internal scores are translated into 11 venue forms: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, AAAI, ACM MM, TMLR, IEEE TPAMI and IEEE TIP. Each translation applies the venue's emphasis weights, bar offset and scale. Each scale is labeled *official*, *confirmed*, *secondary*, *assumed* or *unpublished*. For ICLR 2026 the Overall maps to the rating set $\{0, 2, 4, 6, 8, 10\}$ as $r = 2(x-1)$. The legacy (2025) set is $\{1, 3, 5, 6, 8, 10\}$.

3 Safeguards

- **Prompt injection.** The manuscript is treated as data, never as instructions. Injected text is reported to the editor as an integrity concern and never raises a score.
- **Self-leak in literature search.** A fingerprint filter refuses queries that could find the reviewed paper. It drops the paper's own arXiv versions, code pages, reviews and acceptance lists unread, and labels every other result as prior (peer-reviewed or 6+ months old), recent preprint, concurrent or newer. Only prior work can be a required comparison. Each run logs the mode (`on`, `strict` or `off`). In `strict` mode a cutoff date applies; in evaluation this is the submission deadline.
- **Recognition.** Reviewers report when they recognize a paper from prior knowledge, and such papers are reported separately.
- **Integrity.** Reviewer personas are fictional, no citations may be fabricated, and every point is tied to a location in the manuscript.

4 Evaluation I: accepted papers and planted flaws

An earlier release of the skill (before 2.10) reviewed 30 recently accepted papers: 10 each from CVPR, ECCV and ICLR, spread over 10 topics. All runs completed with every required section; the median review length was 10,214 words. Table 2 reports grounding.

Planted flaws. In six accepted papers we inserted one sentence admitting a fatal evaluation flaw: hyperparameters tuned on the test set, or the test split merged into training.

- The reviews mentioned the flaw in 6/6 papers.
- The Devil's Advocate raised it as DA-CRITICAL in 2/6, and the EIC upheld both.
- The final decision was ≤ 3 in 5/6.

The miss was an ICLR paper rated 4.

Decisions. 16/30 accepted papers landed on the accept side (Overall ≥ 4); 14 received 3. This compression motivated the pool-position calibration of version 2.10. The DA raised CRITICAL issues on 11/30 papers, and the EIC upheld none on real papers.

5 Evaluation II: ICLR 2026 calibration

5.1 Protocol

We follow the paperreview.ai protocol [5].

- Sample ICLR 2026 submissions with human reviews, including withdrawn ones.
- Fit an ordinary least-squares map from the seven dimension scores to the human rating on a train split.
- Report on the test split: Spearman with one human rating, and the AUC of the score for acceptance. Confidence intervals come from bootstrap resampling; results are given for all scored papers and without withdrawn papers.

The harness (`evals/iclr_calibration.py`, standard library only) runs in GitHub Actions through the API. Reviewers never see human scores or decisions.

Error metrics. Following the CSPaper benchmark [6], for predictions $\hat{s}_j$ and human ratings s_j on a scale $[S_{\min}, S_{\max}]$ (ICLR 2026: $[0, 10]$),

$$\mathrm{NMAE} = \frac{1}{n} \sum_j \frac{|s_j - \hat{s}_j|}{S_{\max} - S_{\min}}, \qquad \mathrm{NME} = \frac{1}{n} \sum_j \frac{\hat{s}_j - s_j}{S_{\max} - S_{\min}}. \tag{2}$$

We report each with its per-paper standard deviation and a bootstrap 95% CI. Pilots are too small to fit the regression, so the `pilot` command scores the skill's own outputs as they are, without any fitting to human scores:

- the *panel rating*: the reviewers' mean Overall, mapped to the ICLR scale by $2(x - 1)$;
- the *final rating*: the integer decision on the ICLR scale.

5.2 Data and a decision leak

The sample is ICLR 2026 with seed 2025: 300 papers drawn, 295 with human scores. Our first pilot used the PDFs from paperreview.ai's dump. These turned out to leak the decision: every accepted paper was a camera-ready version (author names, acknowledgments, no line numbers), while all other papers were anonymous submissions. Overall, 0/29 accepted papers and 56/56 other papers showed "Anonymous authors". We therefore use only arXiv versions posted before the 2026-01-26 decision date, normalized identically. The header before the abstract is replaced by the title, and line numbers, arXiv stamps, emails, affiliation footnotes, code links, acknowledgments and acceptance marks are removed. The literature check runs in `strict` mode with the submission deadline as cutoff.

The pilot set has 20 papers stratified by decision and human score: 5 accepted, 10 rejected, 4 withdrawn and 1 desk rejected.

5.3 Pilots

Pilot	Skill	Model / effort	Outcome
1	2.9	Haiku / default	decision leak in the text; scores compressed; Spearman −0.12, AUC 0.47
2	2.10	Haiku / default	normalized text; Spearman −0.01, AUC 0.49 (mean of 6 dimensions)
3	2.10	Sonnet / high	normalized text; Tables 3–4
4	2.11	Sonnet / high	redesigned panel; Spearman 0.32, AUC 0.55; withdrawn (Section 5.7)
Full	2.10.1	Sonnet / high	balanced set, 86 papers; Section 5.5

Table 3: Ranking on the 20-paper pilot: Spearman with the mean human rating [bootstrap 95% CI] / acceptance AUC. Human references are on the same papers unless noted.

Run	Panel rating (1–6)	EIC pool position	Mean of 6 dimensions
Haiku, default	+0.20 [−0.21, 0.58] / 0.56	−0.06 [−0.55, 0.42] / 0.42	−0.01 [−0.52, 0.49] / 0.49
Sonnet, high	+**0.48** [0.09, 0.74] / **0.67**	+0.30 [−0.14, 0.69] / 0.55	+0.22 [−0.27, 0.64] / 0.57
Without the 3 papers Sonnet recognized ($n = 17$):			
Haiku, default	+0.07 / 0.50	−0.25 / 0.33	−0.10 / 0.44
Sonnet, high	+0.46 [0.09, 0.74] / 0.69	+0.22 / 0.53	+0.10 / 0.54
References: one human vs another +0.12, one-human AUC 0.58 (same papers); our ICLR 2026 test split 0.25 / 0.75; paperreview.ai (ICLR 2025) AI 0.42 / 0.75, human 0.41 / 0.84.			

Table 4: Error on the ICLR 2026 scale (0–10) against the mean human rating, 20 papers. NMAE ± std [95% CI]; NME ± std.

Predictor	NMAE ↓	NME	Spearman	AUC
Haiku, panel rating	0.093 ± 0.057 [0.068, 0.118]	$+0.030 \pm 0.105$	+0.21	0.56
Haiku, final rating	0.085 ± 0.063 [0.058, 0.115]	-0.020 ± 0.104	n/a (constant)	0.50
Sonnet, panel rating	$\mathbf{0.079} \pm 0.061$ [0.054, 0.106]	$\mathbf{+0.004} \pm 0.100$	**+0.48**	**0.67**
Sonnet, final rating	0.090 ± 0.064 [0.061, 0.119]	$+0.010 \pm 0.110$	+0.26	0.53
Constant at the median human mean	0.085 ± 0.060	–	–	–
One human vs another human	0.180 ± 0.125	–	+0.12	–
One human vs mean human	0.102 ± 0.065	–	–	–

In pilot 1 nearly every paper received Overall 3, including papers with human means of 2.0 and 6.0. The anchors described papers only relative to *accepted* papers, and the anti-harshness rule could only raise scores. Version 2.10 added pool positions, base rates, two-sided checks and EIC rulings on borderline panels.

5.4 Findings

- **Model capability matters.** Haiku's scores did not separate papers. Its integer final rating was the same for all 20 papers, so Spearman is undefined. Sonnet at high effort gave the first signal whose CI excludes zero, at about the level of human–human agreement.
- **NMAE rewards the middle.** An oracle constant at the median human mean scores 0.085. That beats one human against another (0.180), and Haiku's constant output scores the same. NMAE must therefore be read together with a ranking metric. Only Sonnet's panel rating beats the constant *and* ranks papers.
- **No bias after calibration.** NME is close to zero for both models, so the 2.10 bands no longer shift scores up or down on this pool.
- **The integer decision loses signal.** Rounding the panel mean to an integer decision lowers Spearman from 0.48 to 0.26 for Sonnet. For ranking, the continuous panel rating should be the primary score.
- **Recognition.** The three papers Sonnet recognized all scored high (EIC pool position 65–70) regardless of outcome. Removing them changes the Sonnet results little.
- **Cost.** About $0.86 per paper for Haiku and $1.97 per paper for Sonnet at high effort.

For reference only: CSPaper reports NMAE 0.128, NME −0.015 and SRC 0.658 for Claude Sonnet 5.5 on ICLR main. That benchmark's sample size and paper overlap are unknown, so the numbers are not directly comparable with ours.

Table 5: Full evaluation (Sonnet, high effort, 86 papers): the panel rating against the mean human rating, Spearman [bootstrap 95% CI] / acceptance AUC, and errors on the ICLR scale.

Papers	n	Spearman	AUC
All	86	+**0.47** [0.27, 0.64]	0.64
Pilot 3's papers	20	+0.48 [0.10, 0.74]	0.67
The 66 new papers	66	+0.48 [0.25, 0.66]	0.64
Not recognized by the reviewer	75	+0.48 [0.28, 0.65]	0.66
Without withdrawn papers	69	+0.40 [0.17, 0.59]	0.61
Test split, against one human (protocol)	41	+0.45 [0.14, 0.69]	0.67
Protocol regression on 7 dimensions, test split	41	+0.19 [−0.13, 0.48]	0.67
References on the same 86 papers: one human vs another +0.14, one-human AUC 0.64; paperreview.ai (ICLR 2025): AI 0.42 / 0.75, human 0.41 / 0.84.			
Error (86 papers)		NMAE ↓	NME
Panel rating		**0.084** ± 0.063 [0.071, 0.098]	−0.003 ± 0.105
Final rating (integer)		0.087 ± 0.064 [0.074, 0.102]	−0.018 ± 0.107
Constant at the median human mean		0.089 ± 0.074	–
One human vs mean human		0.118 ± 0.087	–
One human vs another human		0.188 ± 0.157	–

5.5 Full evaluation: 86 papers

The balanced set of the protocol has 86 papers (45 train, 41 test; 23 accepted, 43 rejected, 17 withdrawn, 3 desk rejected). We reviewed the 66 papers not in the pilot with skill 2.10.1, which keeps the 2.10 panel, scoring and decision, and reused the 20 pilot 3 runs. All 86 runs finished, at $166.61 in total ($1.94 per paper). Table 5 gives the results.

- **The pilot signal holds.** The 66 new papers rank exactly as well as the 20 pilot papers (0.48), and recognized papers do not drive the result: the reviewer reported knowing 11 of the 86, and without them Spearman is still 0.48.
- **Above human–human agreement, at paperreview.ai's level.** The panel ranks papers far more like the human mean than one human ranks like another (0.47 vs 0.14), and on the test split against one human (0.45) it is at paperreview.ai's AI level (0.42, a different year and sample). Acceptance AUC (0.64) equals one human's here but stays below paperreview.ai's 0.75.
- **The fitted regression does worse than no fitting.** The protocol's least-squares map from seven dimensions, fitted on 45 papers, reaches only 0.19 on the test split; three of its seven weights are negative. The unfitted panel rating is the better score.
- **No stage loses signal.** The EIC-weighted rating (0.45), the reviewers' pool position (0.48) and the panel without the Devil's Advocate (0.48) are all within ±0.02 of the panel rating, with intervals that include zero.
- **Compression.** The final Overall is 3 for 67 of 86 papers. The mean panel rating (ICLR scale) is 4.45 for accepted papers against a human 5.28, and 4.10 for rejected papers against 3.89: the order is right, but the gap is 0.35 points instead of 1.39.
- **Withdrawn papers help.** Without them Spearman drops to 0.40; part of the signal is spotting clearly weak submissions.

The reviewer's workspace contained no human scores for the 66 new papers; the workflow now also removes all evaluation data from it before each review.

5.6 Issue overlap and hit rate

Scores say nothing about *which* weaknesses a review raises. On the 86 papers of the full evaluation (Sonnet, high effort, skill 2.10.1) we compared the weakness sections of the AI panel with those of the ICLR 2026 human reviews (3–5 per paper). For each paper, one Sonnet annotator saw both panels under random names, in random order and without roles. It split them into atomic

points, grouped the points into issues within each panel and matched issues across panels; it did not judge correctness and never saw scores or decisions. We count main issues only (minor and cosmetic points excluded) and give 95% bootstrap intervals over papers. The method was set on pilot 3's 20 papers, where four papers annotated twice by independent annotators differ by at most 0.04 on every metric; every pilot hit rate is reproduced within 0.06 on the 66 new papers. The tool and the annotation prompts are in the repository (`evals/issue_overlap.py`).

Table 6: Issue overlap and hit rate, 86 ICLR 2026 papers, main issues (pilot 3's 20 papers in the last column).

	Human	AI, 5 reviewers	AI, R1–R4
Main issues per paper	11.8 [11.1, 12.6]	20.4 [19.5, 21.2]	–
One reviewer only	0.77 [0.74, 0.80]	0.48 [0.46, 0.50]	0.73 [0.71, 0.76]
Pairwise Jaccard	0.11 [0.09, 0.13]	0.15 [0.14, 0.16]	0.09 [0.08, 0.10]
Hit rate	86 papers		Pilot (20)
Human issues hit by the AI panel	**0.46** [0.42, 0.49]		0.45
raised by 2+ humans	**0.74** [0.67, 0.80]		0.78
raised by one human	0.38 [0.34, 0.42]		0.38
AI issues also raised by humans	0.26 [0.24, 0.28]		0.26
One human's issues, by other humans	0.41 [0.36, 0.45]		0.39
One human's issues, by 1 / 3 / 5 AI reviewers	0.21 / 0.42 / 0.52		0.20 / 0.40 / 0.50

Table 7: Per-role contribution, 86 papers (means per paper).

Role	Issues	Raised by no other role	Human issues hit	Hit by no other role
R1 Methodology	9.9	0.25	0.22	0.02
R2 Domain	5.4	0.51	0.18	0.08
R3 Perspective	5.2	0.58	0.15	0.06
R4 Generalist	4.8	0.21	0.14	0.02
Devil's Advocate	8.9	0.07	0.21	0.01

- **Redundancy comes from two roles.** The five-reviewer panel repeats itself more than a human panel, but at equal size (R1–R4) its overlap matches the humans' (Jaccard 0.09 vs 0.11). The Devil's Advocate raises 93% of its issues together with another role, and R4 79%.
- **The panel finds what humans agree on.** It raises 46% of the humans' main issues and 74% of those two or more humans share. Three AI reviewers anticipate a given human's issues as well as the other humans do (0.42 vs 0.41). Coverage grows with panel size (0.18, 0.30, 0.37, 0.42, 0.46 for 1–5 reviewers).
- **Too many issues.** The panel raises 1.7× as many main issues, and a quarter are also raised by a human. Unmatched issues are not necessarily wrong, but the reader needs a short list.
- **Votes and issues differ.** On the pilot, R4 added few issues, yet the Spearman of the mean Overall dropped without it (R1–R4 0.52, R1–R3 0.48). Version 2.11 tested a panel that keeps R4's vote but limits its weaknesses, moves the Devil's Advocate into an independent Report Checker and adds 3–5 key concerns (Section 5.7).

Importance. The human reviews carry no severity labels, so we use three proxies: how many humans raised the issue, the rating of the reviewers who raised it (on the ICLR 2026 scale, 4 or less is a reject vote) and whether it was among the first two main points of a review (Table 8). Every proxy points the same way: the more important the issue, the more often the panel raises it, from 0.29 for a single accept-voter's later point to 0.74 for consensus issues. Of the hit issues, 63% of the consensus ones and 56% of the others match an AI issue raised by two or more AI reviewers, so the panel's own agreement tracks importance only weakly. By decision, the hit rate

is lower on accepted papers (0.39 [0.32, 0.45], 23 papers) than on rejected ones (0.49 [0.45, 0.54], 43 papers): their reviewers raise fewer issues (11.0 per paper), and fewer of the AI's issues match theirs (0.19 vs 0.29).

Table 8: Hit rate of the 1017 human main issues by importance proxy, 86 papers.

Human main issues	n	Hit rate [95% CI]
Raised by 2+ humans	218	**0.74** [0.67, 0.80]
Raised by one human	799	0.37 [0.34, 0.40]
Raised by a reviewer who voted reject (≤ 4)	789	0.48 [0.45, 0.52]
Raised only by reviewers who voted accept (≥ 6)	228	0.34 [0.28, 0.40]
Raised by the paper's lowest-rating reviewer	554	0.50 [0.46, 0.54]
Among a reviewer's first two main points	535	0.51 [0.47, 0.56]
Not among any reviewer's first two	482	0.38 [0.34, 0.42]
One human, accept votes only, not a lead point	91	0.29 [0.20, 0.38]

What the panel misses. Of the 1018 human main issues, 456 were hit, 460 had only a related AI issue and 101 had nothing close. A Sonnet labeler (one per five papers) gave each issue a topic and the evidence needed to raise it (before seeing which were hit) and, for the 561 non-hits, the likely reason, by comparing it with the AI issue summaries (Table 9). 82% of the non-hits have a related AI issue: 244 are issues the panel raised only as a broad category ("more baselines") while the human named the item (a specific method or dataset), and 110 more are related concerns with a different diagnosis. True misses are mostly subjective scope requests (58) and fine technical detail (21). Method clarity (0.22), presentation (0.20) and scope (0.33) are hit least; missing experiments, the largest topic (28% of human issues), are hit 0.47. Issues that need outside knowledge are hit *more* often than text-only ones (0.51 vs 0.44), so the literature check is not the bottleneck; figure-dependent issues are hit least (0.29) but are rare (45).

Table 9: Hit rate of the 1018 human main issues by topic and evidence needed, and the likely reason for the 561 non-hits (related: a partial AI overlap; missed: nothing close), 86 papers.

Topic	n	Hit	Reason (non-hits)	Related	Missed
Missing experiments	287	0.47	Raised only as a broad category	242	2
Scope, limitations	148	0.33	Different diagnosis	107	3
Method clarity	107	0.22	Subjective request	59	58
Novelty, related work	100	0.60	Fine technical detail	19	21
Baselines	99	0.63	Needs a specific prior work	14	8
Unsupported claims	91	0.37	Needs figures	11	6
Theory	81	0.51	Unclear	8	3
Efficiency	50	0.64			
Reproducibility	28	0.43			
Presentation	20	0.20			
Evidence needed					
Text and tables	721	0.44			
Outside knowledge	200	0.51			
Derivation	52	0.40			
Figures	45	0.29			

Raising the hit count: a Gap Scout (pilot, 20 papers). Since most non-hits were raised only as a category, we tested one extra, unscored pass: a Sonnet agent that reads only the redacted paper and lists up to 30 concrete weaknesses reviewers would likely raise (claims and their missing evidence, named baselines, undefined elements, theory, scope, typical requests). One annotator

per paper then matched each human main issue against the panel's issues and the Scout's items, mixed without sources, so both are judged on the same scale; the panel's re-judged hits agree with the original annotation on 88% of human issues. Table 10 gives the results. Twenty concrete items hit as many human issues as the whole panel; thirty hit 1.4× as many, and together with the panel 73%. The gain is largest for missing experiments (0.35 → 0.76) and baselines (0.55 → 0.82). Part of it is volume (about 51 items per paper against 12 human issues), and the Scout's items were not checked against the paper in this test. Version 2.10.1 offers the Scout as an optional post-review checklist of anticipated reviewer concerns, outside the review (Section 2).

Table 10: Human main issues hit (247 issues, 20 papers), judged by one annotator per paper.

Source	Items/paper	Issues hit	Hit rate [95% CI]
Five-reviewer panel	20.9	110	0.47 [0.38, 0.55]
Gap Scout, first 10 items	10	70	0.30 [0.23, 0.39]
Gap Scout, first 20 items	20	108	0.46 [0.39, 0.54]
Gap Scout, all	30	153	**0.65** [0.59, 0.71]
Panel + Gap Scout	50.9	175	**0.73** [0.68, 0.79]

Caveats: one LLM annotator and one labeler per paper; the Gap Scout test covers only the 20 pilot papers; importance is inferred from the human reviews, not judged; human reviewers saw the figures, the AI read the extracted text; agreement with humans is not correctness.

5.7 Pilot 4: a redesign that did not help

Version 2.11 put the findings above into the review itself: four voting reviewers (R1 and R2 asked to name concrete items, R4 limited to presentation and venue fit), the Gap Scout as a sixth, unscored pass, and a fresh-context Report Checker that ran the literature and report checks, the Devil's Advocate challenge, the key concerns, base reviewer weights and its own independent estimate. We ran it on the same 20 papers (Sonnet, high effort) and annotated hits on the same 247 human main issues, with one fresh annotator per paper (Table 11).

Table 11: Pilot 4 (skill 2.11) against pilot 3 (skill 2.10), same 20 papers. Hit rate of the 247 human main issues [95% CI]; ranking against the mean human rating.

	Pilot 3 (2.10)	Pilot 4 (2.11)
Panel hit rate	0.47 (five reviewers)	0.43 [0.35, 0.53] (four)
Gap Scout hit rate	0.65	0.65 [0.58, 0.72]
Panel + Gap Scout	0.73	0.74 [0.68, 0.80]
Devil's Advocate hit rate	–	0.01 (in the checker)
Panel rating: Spearman / AUC	**0.48** / **0.67**	0.32 [−0.19, 0.74] / 0.55
Panel rating: NMAE / NME	**0.079** / +0.004	0.083 / −0.028
Final integer rating: Spearman / AUC	0.26 / 0.53	0.24 / 0.47
Checker's own estimate: Spearman / AUC	–	0.51 [0.15, 0.77] / 0.67
Cost per paper	$1.97	$6.93

- **No gain in hits.** The named-item instructions did not raise the panel's hit rate; the additional hits still came from the Gap Scout (75 human issues no other source raised).
- **Worse ranking, compressed ratings.** R1 gave 2 on 13 papers and R4 gave 3 on 17; the paired difference of the panel rating's Spearman, pilot 4 − pilot 3, is −0.15 [−0.48, 0.17]. A plausible reading is that the more detailed instructions made the reports finer but their scores less aligned with the human ones.
- **The checker's estimate** ranked at 0.51, the only interval in pilot 4 that excludes zero, but it used three Overall values and is no better than the 2.10 panel's 0.48 within the error bars

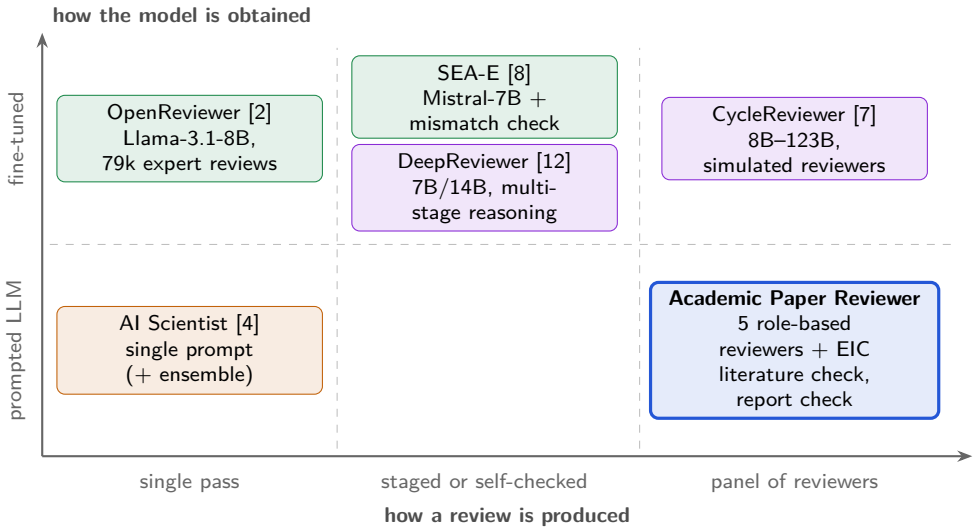


Figure 2: Design space of AI reviewers. Fine-tuned systems learn from public reviews of past conferences. Our skill prompts a strong general model through a structured panel and needs no training data. Positions are qualitative.

(difference to the 2.11 panel 0.19 [−0.32, 0.70]).
Version 2.10.1 therefore returns the panel, its instructions, the scoring and the decision rule to 2.10, keeps the Gap Scout only as an optional post-review checklist, and removes the Report Checker.

6 Related work

AI reviewer systems. Figure 2 places our system among published AI reviewers along two axes: how the model is obtained and how a review is produced.

- **Prompted general models.** The AI Scientist [4] reviews with one prompted pass, optionally with reflection and an ensemble.
- **Fine-tuned single pass.** OpenReviewer [2] fine-tunes Llama-3.1-8B on about 79k expert reviews; its ratings follow the human distribution, while general models rate too leniently.
- **Fine-tuned with a consistency check.** SEA [8] standardizes multiple reviews into one format (SEA-S), fine-tunes Mistral-7B on them (SEA-E), and regenerates a review when a mismatch score (SEA-A) finds it inconsistent with the paper.
- **Fine-tuned with simulated reviewers.** CycleReviewer [7] (8B–123B) simulates several reviewers and a senior reviewer and averages their scores. It is trained on ICLR 2024 reviews and serves as the reward model of CycleResearcher.
- **Fine-tuned with multi-stage reasoning.** DeepReviewer [12] (7B, 14B) is trained on DeepReview-13K, synthetic stepwise reviews of ICLR 2024–2025 papers, and reasons in stages with Fast, Standard and Best modes.

Our system is closest in spirit to CycleReviewer and DeepReviewer: a panel of perspectives and staged reasoning. It differs in four ways:

- it is a prompt-only skill for a strong general model, so it needs no training data and can use a newer model as soon as one is released;
- the reviewers have distinct roles, including an adversarial Devil's Advocate;
- an explicit EIC weights reviews by quality instead of averaging them;
- scoring is deterministic and venue-specific.

Table 12: Published rating agreement with the mean human rating (1–10 scale). Values are from Zhu et al. [12], Table 2 (columns ICLR 2024 / ICLR 2025). Last row: our pilot.

System	Spearman	Rating MAE	Decision acc.
DeepReviewer-14B [12]	0.356 / 0.405	0.91 / 0.92	0.64 / 0.69
CycleReviewer-70B [7]	0.336 / 0.267	1.25 / 1.21	0.63 / 0.68
CycleReviewer-8B [7]	0.280 / 0.279	1.24 / 1.21	0.64 / 0.68
AI Scientist, DeepSeek-R1 [4]	0.326 / 0.324	1.65 / 1.81	0.52 / 0.43
AI Scientist, Claude 3.5 Sonnet [4]	0.037 / −0.022	1.50 / 1.35	0.48 / 0.56
Ours, Sonnet high, panel rating (ICLR 2026, $n = 20$)	0.48	0.79†	–

† On the ICLR 2026 0–10 scale (NMAE 0.079×10).

Reported scores of reviewer models. Table 12 collects published score-agreement numbers. They come from different paper sets, sizes and targets and are *not* comparable with our 20-paper pilot. Because all the fine-tuned models were trained on data up to ICLR 2024–2025, ICLR 2026 is unseen by them, which makes them fair open-weight baselines for our sample (future work).

Other studies report MAE 0.86 for OpenReviewer-8B and 1.10 for SEA-E under their own setup [3]. NAIPv2 [11] could not reproduce these systems and quotes their published numbers. Re^2 [10] evaluates all four fine-tuned systems on its review and rebuttal data.

Benchmarks.

paperreview.ai [5]. Score agreement and acceptance AUC on ICLR; our protocol follows it.

CSPaper [6]. A score-prediction leaderboard with NMAE, NME, SRC and average word count; we adopt its error metrics.

CoCoReviewBench [1]. This benchmark scores review *content* rather than scores. It covers 3,900 ICLR 2017–2025 and NeurIPS 2021–2024 papers, with human references filtered for correctness through reviewer–reviewer and reviewer–author conflicts, using the meta-review as context. An LLM judge scores correctness, thoroughness, grounding, verifiability and clarity per category, and the benchmark also reports category completeness. Among its findings: lexical metrics reward overlap rather than correctness, reasoning models (including DeepReviewer-14B) match or beat humans on grounding and verifiability, and AI reviews are broad but shallow.

AI-text detection [9]. About 789k paired human and AI reviews for *detecting* AI-written reviews. It has a different goal and a restrictive license, so we do not use it.

7 Limitations and future work

- **Sample size.** The full evaluation has 86 papers; its Spearman interval is still 0.27–0.64, and the issue-overlap study and the 2.11 comparison use the 20-paper pilot only. Haiku was tested on the pilot only.
- **One venue and one year.** Calibration covers ICLR 2026 only. Other venues use assumed or secondary scales.
- **Memorization.** Strong models recognize some papers. We report recognized papers separately but cannot detect unreported recognition.
- **Scores are not the review.** Ranking agreement says nothing about whether the stated weaknesses are correct, and the overlap study (Section 5.6) measures agreement with humans, not correctness. A content-level evaluation (for example CoCoReviewBench [1]) is the natural next step, together with open-weight baselines [7, 12] run on the same ICLR 2026 papers.
- **The Gap Scout's items are unchecked.** Its list is longer than a human panel's, and part of its hit rate is volume; how many of its items are wrong or already answered by the paper was not measured.

- **Compressed ratings.** On the 86 papers, 78% receive final Overall 3 (ICLR 4); accepted papers are rated 0.8 points below the humans. Better ranking at the top and bottom of the pool is the main open problem.
- **Human labels are noisy.** On the 86 papers, one human agrees with another at only Spearman 0.14. Mean human ratings are a weak target, and acceptance is a coarse one.

Reproducibility

The skill, the scoring script, the evaluation harness, per-paper scores and paper IDs are in the repository. Manuscripts and review text are not committed. Key commands:

```
python3 -m unittest discover -s tests -v
python3 evals/iclr_calibration.py features RESULTS_DIR --out features.jsonl
python3 evals/iclr_calibration.py pilot sample.jsonl features.jsonl
python3 evals/iclr_calibration.py evaluate sample.jsonl features.jsonl \
        --splits splits.json
```

Paid runs use the workflow `.github/workflows/iclr-calibration.yml`, configured by `evals/iclr_run.json`.

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A Per-paper pilot scores

		Human	Haiku		Sonnet high		
Paper	Decision	mean	panel	pool	panel	pool	Recognized
AIKURJjxXT	Withdrawn	2.50	3.2	40	2.8	40	
KtocialdxS	Reject	2.67	3.4	48	3.0	45	
8xcmfwomnI	Reject	3.00	3.0	40	3.0	42	
g4HpTtgsn2	Withdrawn	3.00	3.2	46	2.6	38	
E5mLO7Fbq8	Reject	3.33	3.0	38	2.6	42	
Z6z4AnXGhw	Reject	3.50	3.4	50	3.2	50	
gUw8n109ki	Withdrawn	3.50	3.2	50	2.6	40	
0xObmDmsKY	Reject	4.00	3.2	46	3.0	45	
BxeJLOrKDF	Withdrawn	4.00	3.2	45	3.2	45	
Jw7khYzYzl	Reject	4.00	3.6	45	4.0	70	✓
9jcB40wjk3	Accept	4.40	2.8	40	2.8	38	
be4ey66fhk	Reject	4.40	3.2	45	3.4	52	
9R2iUHhVfr	Reject	4.50	3.4	50	3.6	65	✓
PJdMrK79Mo	Accept	4.50	3.6	48	3.4	50	
qfW4IrUuig	Reject	4.50	3.0	40	2.8	40	
LvkbeOCgmZ	Reject	5.00	3.4	40	3.2	45	
tlYSbw5GXY	Accept	5.20	3.2	38	3.4	55	
euMVC1DO4k	Accept	6.00	3.6	50	3.8	68	✓
kwro5432AI	Desk rej.	6.00	3.2	45	3.0	42	
rubeJmT1XM	Accept	6.00	3.2	40	3.0	40	

Human mean is on the ICLR 2026 scale (0–10). “Panel” is the reviewers’ mean Overall (1–6) and “pool” is the EIC pool position (0–100).